FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Genital Prolapse

Tutorial | Week 4 · Hypertension, twins & the pelvic floor | Gynaecology | Dr Nassimu

Pelvic organ prolapse (POP) is the descent of the uterus, the vaginal vault or the vaginal walls (with the bladder, rectum or bowel behind them) because the pelvic floor muscles and fascial supports have failed, most often after vaginal childbirth and menopause. It is graded with POP-Q (stages 0–IV) or Baden–Walker. It is treated with lifestyle measures, pelvic floor muscle training and pessaries, or surgery (vaginal hysterectomy with vault support, Manchester repair, anterior or posterior colporrhaphy, sacrospinous fixation or sacrocolpopexy), chosen to suit the woman's symptoms, fitness and wishes.

Must know before the ward round

1. Support has three levels (DeLancey). Level I (uterosacral–cardinal complex) suspends the apex, Level II (paracolpium to the arcus tendineus) attaches the mid-vagina, Level III (perineal body, perineal membrane) fuses the distal vagina. Level I failure causes uterine or vault prolapse.
2. The strongest risk factor is vaginal parity (especially large babies, a long second stage, instruments). Others: age, menopause, obesity, chronic cough, constipation, heavy lifting, connective tissue disease, previous hysterectomy (vault).
3. POP-Q uses the hymen as 0. Stage I: leading edge more than 1 cm above the hymen. Stage II: within 1 cm of the hymen. Stage III: more than 1 cm beyond the hymen but less than complete. Stage IV: complete eversion (procidentia).
4. Examine with a Sims speculum in the left lateral position, with the woman straining. Retract the posterior wall to see a cystocele and the anterior wall to see a rectocele or enterocele. Do a rectovaginal exam to tell a rectocele from an enterocele.
5. Treat only symptomatic prolapse. First line is supervised pelvic floor muscle training for at least 16 weeks (NICE) plus weight loss and treating cough or constipation. Pessary (ring or Gellhorn) for women who decline or are unfit for surgery.
6. Manchester (Fothergill) repair = cervical amputation + cardinal ligament shortening + anterior repair. It preserves the uterus, but risks cervical stenosis and incompetence, so it suits women who have completed their family.
7. Procidentia with a decubitus ulcer: admit, reduce the prolapse and keep it reduced with a daily vaginal pack (oestrogen cream where available), treat infection, check for hydronephrosis, and biopsy any ulcer that does not heal, before definitive surgery.
8. NICE advises against transvaginal mesh for anterior or posterior wall repair outside research. Sacrocolpopexy (mesh via the abdomen) is the most durable operation for vault prolapse.
9. Record POP-Q as a 3 × 3 grid (Aa, Ba, C / gh, pb, tvl / Ap, Bp, D) in cm from the hymen on maximum strain; only tvl is measured at rest.
10. In every advanced prolapse check the post-void residual (over about 100–150 mL is incomplete emptying), U&E and a renal ultrasound for hydronephrosis before planning surgery.
11. Reduce the prolapse and ask her to cough before surgery: a leak means occult stress incontinence that the repair may unmask.

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1. Definition

- **Genital (pelvic organ) prolapse** is the **downward displacement of the pelvic organs into or out of the vagina**. The uterus and cervix, the **vaginal vault** (after hysterectomy), or the anterior or posterior vaginal walls can come down, carrying the **bladder (cystocele), urethra (urethrocele), small bowel (enterocele) or rectum (rectocele)** with them.
- It is a **disease of failed support**. Vaginal childbirth tears and stretches the levator ani and its fascia and injures the pudendal nerve. Menopause and ageing weaken the collagen. Chronically raised intra-abdominal pressure (**cough, constipation, heavy loads**) keeps pushing down.
- It is **rarely life-threatening but very disabling**: dragging discomfort, a bulge, difficulty passing urine or stool, and sexual and social problems.

Term	Meaning
Cystocele	The bladder bulges through the upper two-thirds of the **anterior** vaginal wall
Urethrocele	The urethra bulges through the lower third of the anterior wall (often with stress incontinence)
Cystourethrocele	Both together (the common presentation)
Uterine (uterovaginal) prolapse	The uterus and cervix descend along the vaginal axis
Procidentia	The **entire uterus lies outside the introitus**, with the vagina everted (POP-Q stage IV)
Vault prolapse	Descent of the vaginal apex **after hysterectomy**
Enterocele	Herniation of the pouch of Douglas (with small bowel or omentum) through the **upper posterior** wall or vault
Rectocele	The rectum bulges through the **lower or mid posterior** wall
Cervical elongation	A long cervix (often supravaginal) that mimics uterine descent while the fundus stays in place

CLINICAL PEARL

The one thing to remember: treat the **woman's symptoms, not the grade**. Always check the **bladder emptying, the kidneys (hydronephrosis in procidentia) and the cervix or ulcer (cancer)** before planning surgery.

2. Epidemiology

- Prolapse is found on examination in **up to about half of parous women**, but only a minority have symptoms (roughly **3–6%** report a bulge). The lifetime risk of having surgery for prolapse or incontinence is about **1 in 10 or more** in high-income countries.
- In Uganda it is common in **grand multiparous, post-menopausal women who do heavy farm work and carry loads**. In **low-income settings** such as Uganda, prolapse is common because of:

- **high parity** (average family size remains around 4–5 children)
- **births at home or without skilled attendants** (long second stages, early pushing, fundal pressure)
- **heavy physical labour** (digging, carrying water, firewood and produce on the head)
- **chronic cough** (TB, smoke from cooking fires, COPD)
- **constipation**
- **poor access to surgery**

- Many women present **late**, with **complete procidentia, decubitus ulcers, urinary retention and hydronephrosis**, often after years of shame, sometimes believing it is a curse or cancer, and some have been rejected by their husbands.

> UGANDA CONTEXT
> Most prolapse surgery in Uganda is **native-tissue vaginal surgery**: vaginal hysterectomy with McCall culdoplasty, anterior and posterior colporrhaphy, and Manchester repair. It is done at regional and national referral hospitals (Mulago, Mengo and others) and during **surgical camps**. **Pessaries** are underused because supply and follow-up are hard, but they are cheap and useful for frail women if follow-up can be assured. **Vaginal oestrogen** is often not on the essential medicines list at lower levels and may have to be bought privately.

3. Aetiology and Risk Factors

Risk factors ("PELVIC SAG")

- **P**arity (vaginal): the strongest risk factor, especially large babies, a long second stage, instruments, premature pushing and fundal pressure.
- **E**lderly: ageing weakens collagen.
- **L**ifting heavy loads (farm work, water, firewood and produce carried on the head).
- **V**aginal instrumental delivery / large babies.
- **I**ntra-abdominal pressure: chronic cough (TB, cooking smoke, COPD), constipation, obesity, ascites, pelvic masses.
- **C**onnective tissue disorders.
- **S**urgery: previous hysterectomy (vault prolapse), especially without vault support.
- **A**trophy (menopause: falling oestrogen weakens collagen).
- **G**enetic / neurological (spina bifida occulta, neurological disease): consider in a young nulliparous woman.

4. Pathophysiology

The pelvic floor: a muscular hammock and a fascial suspension

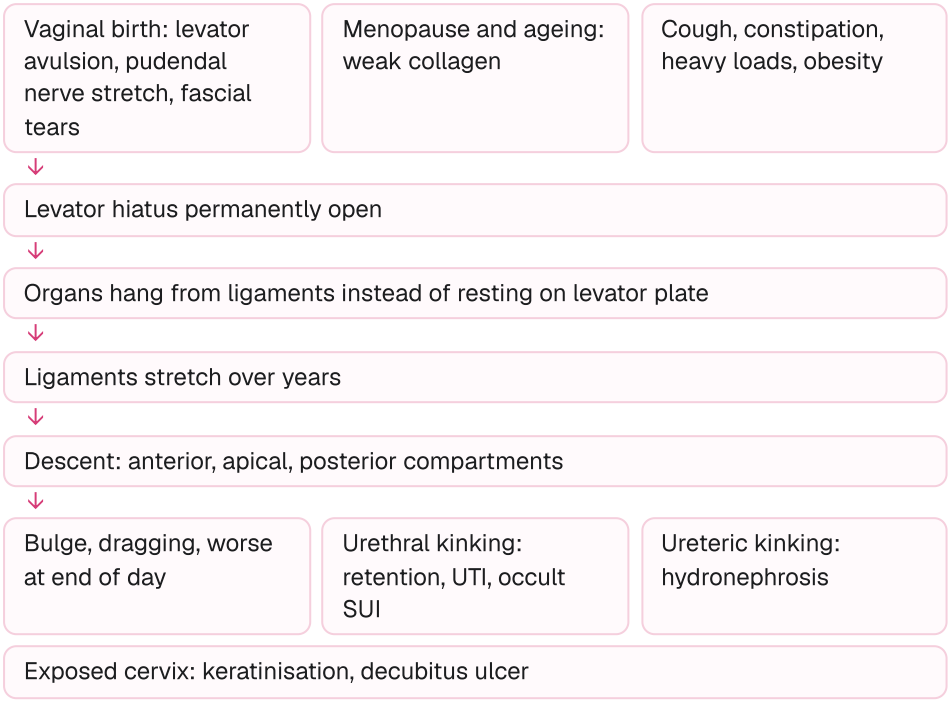
1. **Levator ani** (pubococcygeus, **puborectalis**, iliococcygeus) plus coccygeus. These muscles form the **pelvic diaphragm**. With **constant resting tone** they keep the **levator hiatus** (the gap through which the urethra, vagina and rectum pass) **closed**. They also lift the organs onto a flat **levator plate**.
2. **Endopelvic fascia** and ligaments: the connective tissue that **attaches the vagina and uterus to the pelvic side walls**.
3. **Perineal body and perineal membrane**: the **distal anchor**.

DeLancey's three levels of vaginal support

Level	Structures	What it does	Failure causes
I (suspension)	**Uterosacral–cardinal ligament complex**	Suspends the **cervix and upper vagina (apex)** towards the sacrum	**Uterine or vault prolapse, enterocele**
II (attachment)	**Paracolpium** to the **arcus tendineus fasciae pelvis** (laterally) and the fascia over the levators	Attaches the **mid-vagina** laterally, forming the "hammock" under the bladder	**Cystocele** (paravaginal or central defect) and **rectocele** (mid)
III (fusion)	**Perineal body, perineal membrane**, superficial perineal muscles	Fuses the **distal vagina** to surrounding structures	**Urethrocele, low rectocele, gaping introitus** (perineal deficiency)

Why childbirth causes prolapse

- During crowning, the **levator ani must stretch to more than 3 times its resting length**. **Avulsion** of the puborectalis from the pubic bone is common after difficult births.
- The **pudendal nerve** is stretched, causing **denervation** and weaker muscles.
- The **fascia tears**.
- Once the levator hiatus is **permanently open**, the organs are no longer resting on a muscular shelf. They hang from the ligaments, which then **stretch over years**, especially when **oestrogen falls** after menopause (collagen weakens) and with every cough or heavy load.
- That is why prolapse often **appears decades after** the births that caused it.



Why prolapse causes the symptoms it does (feature links)

- Gravity on stretched ligaments → **bulge and dragging**, worse at the end of the day and on standing, better lying down.
- Large cystocele → **urethral kinking** → hesitancy, incomplete emptying, a need to **reduce the bulge with a finger to void** → residual urine → recurrent UTIs.
- Procidentia → **ureters dragged down** and compressed by the levator hiatus → **hydronephrosis** → obstruction and, rarely, renal failure.
- Prolapse kinks the urethra → masks SUI → **occult stress incontinence** that can **appear after repair**.

- Rectocele → stool pockets into the bulge → **incomplete evacuation** and a need to **splint** (press on the posterior vaginal wall or perineum to defaecate).
- Exposed cervix or vagina → keratinised, dry, friction-damaged; venous congestion and low oestrogen add ischaemia → **decubitus ulcer** (can mimic cancer), bleeding, discharge.
- Level I failure → apical descent (uterus, vault, enterocele); Level II → cystocele and mid rectocele; Level III → urethrocele, low rectocele, gaping introitus.
- Oedema in an irreducible prolapse → venous congestion → irreducibility, rarely strangulation or gangrene.

5. Classification / Staging

Traditional grading of uterine prolapse (still used on Ugandan wards)

Degree	Description
1st	The cervix descends but stays **within the vagina**
2nd	The cervix reaches or protrudes **through the introitus** on straining; the body of the uterus stays inside
3rd (procidentia)	The **whole uterus lies outside** the introitus

Baden–Walker halfway system

Grade 0 = normal position. Grade 1 = halfway to the hymen. Grade 2 = to the hymen. Grade 3 = halfway past the hymen. Grade 4 = maximum descent.

POP-Q (Pelvic Organ Prolapse Quantification)

This is the international standard (ICS/IUGA). The **hymen is the reference point (0)**. Points above it are negative (in cm) and points below it are positive. All are measured at **maximum strain (Valsalva)** except tvl.

Point	Where it is	Normal range
Aa	Anterior wall, **3 cm** above the hymen (near the urethrovesical junction)	−3 (normal) to +3
Ba	Most dependent part of the rest of the anterior wall	−3 (normal) up to +tvl
C	The cervix (or the vault after hysterectomy)	About −tvl (normal) to +tvl
D	Posterior fornix (omitted after hysterectomy)	Deep in the pelvis normally
Ap / Bp	Posterior wall equivalents of Aa and Ba (Ap 3 cm from the hymen, range −3 to +3)	As for Aa / Ba
gh (genital hiatus)	Middle of the urethral meatus to the posterior hymen	cm
pb (perineal body)	Posterior hymen to the mid-anal opening	cm
tvl (total vaginal length)	Depth of the vagina **at rest**, with the prolapse reduced	cm

The 3 × 3 grid: row 1 = Aa, Ba, C (anterior wall and apex); row 2 = gh, pb, tvl; row 3 = Ap, Bp, D (posterior wall and posterior fornix).

Stage	Criterion
0	No prolapse (Aa, Ap, Ba, Bp all −3; C or D within 2 cm of −tvl)
I	Most distal part is **more than 1 cm above** the hymen (less than −1)

Stage	Criterion
II	Most distal part is **within 1 cm of the hymen** (–1 to +1)
III	Most distal part is **more than 1 cm below** the hymen, but no further than (tvl – 2) cm
IV	**Complete eversion** (at least tvl – 2 cm): procidentia or complete vault eversion

> **EXAM FAVOURITE**
> POP-Q is objective and reproducible, which is why it is used in research and guidelines. On a busy Ugandan ward, you will more often hear "**third-degree uterine prolapse with a cystocele and a rectocele**". Know both, and translate: "third degree" roughly equals POP-Q stage III–IV.

By compartment ("Front, top, back"): front = bladder (cystocele, urethrocele); top = uterus, vault or enterocele; back = rectum (rectocele).

6. Clinical Features

Most women with prolapse are worked up calmly in the gynae clinic or on the ward before a surgical camp. The workup answers: **Which compartments are down, and how far (POP-Q or degree)? Is the bladder emptying and are the kidneys safe? Is there an ulcer or bleeding that could be cancer? What does she want (fertility, sex), and is she fit for which treatment?**

History

Ask	Why it changes the plan
The bulge: "something coming down"? How long, worse when standing or at the end of the day, can she push it back, is it always out?	Degree and impact; always out suggests stage III–IV; irreducible suggests oedema or ulceration
Voiding: frequency, hesitancy, poor stream, **pushing the bulge back to void**, incomplete emptying, recurrent UTI, dysuria, loin pain	Cystocele, urethral kinking and retention, UTI, hydronephrosis: check the residual and the kidneys
Leakage on cough or sneeze (and whether it got better as the prolapse got worse)	SUI or **occult SUI**: test with the prolapse reduced; may need a continence procedure
Urgency, frequency, nocturia	OAB or UTI; do not promise that surgery cures urgency
Bowels: constipation, straining, incomplete evacuation, **splinting or digitating**, faecal incontinence	Rectocele, enterocele, sphincter injury: posterior repair is indicated only if symptomatic
Bleeding or blood-stained discharge; postmenopausal bleeding	**Decubitus ulcer** vs **cervical or endometrial cancer**; PMB needs investigation
Sexually active? Dyspareunia? Wishes for the future?	Colpocleisis (no future intercourse) vs reconstructive surgery; avoid levator plication in sexually active women; posterior repair can cause dyspareunia
Wants more children?	Conservative or uterus-preserving surgery; avoid Manchester repair (cervical incompetence) and hysterectomy
Obstetric history: parity, birth weights, home or long births, instruments, tears	Risk factors; explains the mechanism; counselling
Menopause; any oestrogen/HRT use	Atrophy: vaginal oestrogen before a pessary or surgery
Chronic cough (TB, cooking smoke, COPD), constipation, heavy loads, occupation, weight	Modifiable "pushers": fix them or the prolapse recurs

Ask	Why it changes the plan
Comorbidity and medicines (hypertension, diabetes, heart or lung disease, anticoagulants)	Fitness for anaesthesia; favours a pessary or colpocleisis for the frail
Previous hysterectomy or repair	Vault prolapse; recurrence; scarring, adhesions
Can she return for follow-up?	A pessary needs reviews every 3–6 months; a woman who cannot return may be safer with surgery

Examination

Follow the order of Hutchison's Clinical Methods (24th edition).

Introduction, consent and preparation

- Introduce yourself, confirm identity, explain, gain consent; **chaperone** for the pelvic examination.
- **Hand hygiene**; gloves.
- Ask her to **empty her bladder first** (then measure the residual); good light.
- **Position and exposure**: supine for the abdomen; **left lateral** for the Sims speculum; standing if the prolapse is not reproduced lying down.

General inspection and vital signs

BMI, pallor, hydration, frailty and mental state (many of these women are elderly); BP, pulse, temperature (pyelonephritis, sepsis).

Hands, face and neck, chest

Pallor; signs of chronic lung disease; **chest** (cough, wheeze, signs of TB or COPD): treat the cause of raised intra-abdominal pressure.

Abdomen

A **palpable bladder**, **renal angle tenderness**, pelvic or abdominal masses and **ascites** (which can push organs down: secondary prolapse), hernias.

Pelvic examination

Step	What you document
Inspect the vulva at rest, then on strain/cough	Anything visible at the introitus; **atrophy**; ulcer (site, size, edge); **urine leak on cough** (SUI); gaping introitus and perineal body length
Sims speculum, left lateral (a Cusco hides the walls)	Insert it to **retract the posterior wall**, then ask her to strain: shows the **anterior wall (cystocele)** and the **cervix descending**. Withdraw it slowly while she strains to see the **apex** descend. Rotate it to **retract the anterior wall**: shows the **posterior wall (rectocele or enterocele)**
Leading edge of each compartment	In cm relative to the **hymen** (POP-Q) or as a degree
Reduce the prolapse and ask her to cough	**Occult SUI**
Standing examination if the story is bigger than the findings	Reproduces the maximum descent
Bimanual	Uterine size and mobility, fundal level, adnexal masses; **cervical elongation** (fundus high, cervix long)

Step	What you document
Rectovaginal (thumb in the vagina, finger in the rectum)	**Rectocele** (rectum bulging into the vagina, felt with the rectal finger) vs **enterocele** (**bowel between your fingers** in the rectovaginal septum, above the rectocele); anal sphincter tone
Cervix and any ulcer	VIA or cytology; **biopsy** anything hard, raised, friable or not healing
Pelvic floor contraction ("squeeze and lift")	Oxford grade 0–5; baseline for PFMT
POP-Q if trained	3 × 3 grid

Completing the examination

- Measure a **post-void residual** (by catheter or bladder scan).
- Thank her, cover her, hand hygiene, summarise.

CLINICAL PEARL

Reduce the prolapse and ask her to cough (a ring forceps or the posterior blade of the Sims holding the anterior wall up). If urine leaks, she has **occult stress incontinence**, and she needs counselling or a continence procedure at the time of the repair.

Red flags

RED FLAG

- **Irreducible, oedematous or gangrenous procidentia**: admit, reduce and elevate.
- **Urinary retention**, a palpable bladder, rising creatinine: **obstructive uropathy** from ureteric kinking.
- **Fever, loin pain, confusion, reduced urine output, vomiting** in an elderly woman with procidentia: obstructive uropathy or sepsis.
- **Bleeding ulcer, or an ulcer that does not heal** after 2 weeks of reduction and packing: **biopsy for carcinoma** (cervix or vagina).
- **Postmenopausal bleeding** with prolapse: exclude endometrial cancer (ultrasound for endometrial thickness, sampling).
- **Prolapse in a young nulliparous woman**: consider **spina bifida occulta** or neurological disease, or a connective tissue disorder.
- **Prolapse in pregnancy** (rare): risk of cervical oedema, infection and obstructed labour.
- A **mass pushing the uterus down** (ovarian tumour, ascites): treat the cause.

Thinking like a clinician

1. **Match the bulge to the compartment.** Anterior (bladder), apical (uterus, vault, enterocele), posterior (rectum). **Missed apical prolapse is the main cause of recurrence after an anterior repair.** If the apex is down, you must support it (vault suspension or hysterectomy with culdoplasty).
2. **Symptoms matter more than the stage.** A stage II prolapse with no symptoms needs no surgery. A stage II prolapse causing retention does.
3. **Ask "Does she want more children? Is she sexually active? Can she survive an anaesthetic?"** Those three questions choose the operation.
4. **Cervical elongation vs true uterine descent**: if the uterine fundus is in its normal place and the cervix is very long, the problem is **elongation**. Manchester repair (which amputates the cervix) suits this.
5. **Look for the pushers**: chronic cough, constipation, heavy lifting, obesity, pelvic masses. If you don't fix them, the prolapse comes back.

6. **Common student mistakes**:
    - using a Cusco (bivalve) speculum, which hides the walls
    - not examining with the woman straining
    - forgetting the **rectovaginal exam** (enterocele)
    - not checking the **residual urine** or the **kidneys**
    - calling every vaginal mass a prolapse (it could be a **fibroid polyp or chronic inversion**)

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Uterine prolapse	The cervix descends along the vaginal axis on straining; reducible; the uterus is small and mobile	Clinical: Sims speculum, POP-Q
Cervical elongation	A long cervix with the fundus at its normal level	Bimanual exam; uterine sound length (much over 8 cm, mostly cervix); ultrasound
Fibroid polyp (pedunculated submucous fibroid)	Firm mass through the os; the cervix **ring is felt around its stalk**; bleeding	Speculum (cervix identified separately), ultrasound
Chronic uterine inversion	Red, velvety mass with **no os visible**; the fundus is absent on abdominal palpation; a history of a difficult third stage	Ultrasound or MRI
Cervical or vaginal carcinoma	Hard, friable, bleeding ulcer or growth, contact bleeding	**Biopsy**
Gartner's duct cyst / vaginal wall cyst	Smooth, cystic, lateral or anterior-lateral wall; does not change with straining	Clinical, ultrasound
Urethral diverticulum	Tender suburethral swelling, post-void dribbling, pus expressed from the urethra	Ultrasound or MRI
Urethral mucosal prolapse (girls)	Red **doughnut** around the meatus, bleeding	Clinical
Bartholin cyst	Posterolateral vulval swelling at 4 or 8 o'clock	Clinical

8. Clinical Workup

The diagnosis and staging of prolapse are **clinical** (Sims speculum and POP-Q): investigations look for **complications** (retention, hydronephrosis, UTI), **other pathology** (cancer, endometrial disease) and **fitness for surgery**. Availability reflects the source notes Verify *compiled from the source notes, not an MoH list*.

Bedside

Test	What you want	Why	Available at
Post-void residual (in–out catheter or bladder scan, straight after she voids)	**Over about 100–150 mL** = incomplete emptying	Retention from kinking: catheterise, check the kidneys; residual urine causes UTIs	All levels (catheter); bladder scan at hospitals
Urine dipstick	Nitrites, leucocytes, blood, glucose	UTI (send culture, treat before surgery); glucose = undiagnosed diabetes	All levels

Test	What you want	Why	Available at
Cough stress test with the prolapse reduced	Leak from the meatus	**Occult SUI**: counsel, or add a continence procedure	All levels
Pregnancy test (premenopausal women)	hCG	Before a pessary fitting or surgery	All levels
HemoCue / Hb	Anaemia	Ideally Hb above 10 g/dL before surgery	HC III upwards
Random blood sugar	Diabetes	Healing, infection risk	HC III upwards
VIA (visual inspection with acetic acid)	Acetowhite lesion	Cervical screening before hysterectomy or Manchester repair	HC III upwards
HIV RDT (if unknown)	Status	Healing, counselling, ART	All levels
Trial pessary fitting (if she chooses one)	Retained on straining, walking and squatting; a finger fits between ring and wall; she can **void** with it in	Confirms the right size before discharge	Where pessaries are stocked

Basic lab

Test	When	Expected / abnormal finding	How it changes management (and where)
Urine microscopy and culture	Dipstick positive, retention, before surgery	Pyuria, organism	Treat per culture (e.g. nitrofurantoin 100 mg 12-hourly for 5 days) before surgery. Culture at regional/national referral hospitals
Urea, creatinine, electrolytes	Procidentia, retention, hydronephrosis, all older women before surgery	Raised creatinine	**Obstructive uropathy**: reduce and pack or catheterise, then recheck; anaesthetic and drug dosing (avoid NSAIDs and nitrofurantoin). Regional/national referral; limited at HC IV
FBC, blood group and cross-match	Before surgery	Anaemia	Correct anaemia; blood available for hysterectomy

Imaging

Test	When	Expected / abnormal finding	How it changes management (and where)
Renal and bladder ultrasound	Procidentia, high residual, raised creatinine, loin pain	**Hydronephrosis**, small kidneys, bladder stones	Hydronephrosis usually resolves after reduction and surgery; persistent obstruction needs urology. Most HC IVs and hospitals
Pelvic ultrasound	Before surgery; **postmenopausal bleeding**; bimanual abnormal	Fibroids, adnexal mass, **endometrial thickness** (thickened in PMB = sample)	Choice of operation; excludes pathology before colpocleisis
Chest X-ray, ECG	Older women, chronic cough, cardiac history	TB, COPD, cardiac disease	Fitness for anaesthesia; treat the cause of raised intra-abdominal pressure

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management (and where)
Endometrial sampling	PMB, thickened endometrium, before colpocleisis	Hyperplasia or cancer	Cancer changes the whole plan (staging, oncology)
Biopsy of an ulcer or lesion	Ulcer not healed after about 2 weeks of reduction and packing, or looks suspicious	Squamous carcinoma vs decubitus ulcer	Carcinoma = oncology, not prolapse surgery. Histology at regional/national referral hospitals
Cervical screening (VIA / cytology / HPV)	Before hysterectomy or Manchester repair	Pre-invasive disease	Treat first or change the operation
Urodynamics	Before a continence procedure when the diagnosis is unclear (mixed symptoms, previous surgery)	SUI vs detrusor overactivity, voiding dysfunction	Selects who benefits from a continence operation. National referral and a few private centres only
Sputum GeneXpert	Chronic cough	TB	Treat the "pusher" before surgery

Putting it together

Model one-line summary: "A 62-year-old para 8, 12 years postmenopausal, with a **complete procidentia (POP-Q stage IV)**, a 2 × 3 cm decubitus ulcer, a **post-void residual of 250 mL** and **bilateral mild hydronephrosis**, with a chronic cough from cooking smoke."

Problem list

1. Uterovaginal prolapse, POP-Q stage IV (apical, anterior and posterior)
2. Decubitus ulcer (carcinoma not yet excluded)
3. Incomplete bladder emptying with bilateral hydronephrosis
4. Chronic cough (cooking smoke; exclude TB)
5. Atrophy; fitness for surgery to be assessed

Working diagnosis and plan: procidentia with ulcer and obstructive uropathy → admit, reduce and daily oestrogen pack, catheter, urine culture, U&E, sputum for TB, biopsy if the ulcer has not healed by about 2 weeks; then vaginal hysterectomy with McCall culdoplasty and anterior and posterior repair (or colpocleisis if frail and not sexually active).

Model case presentation (ward round)

> "Mrs K is a 62-year-old para 8, post-menopausal for 12 years, a farmer from Mukono. She has a 4-year history of 'something coming out of the vagina', which is now always outside, with a bloody discharge, difficulty passing urine (she has to push the mass back to void) and constipation. She has a chronic cough from cooking smoke. On examination she has a **complete procidentia (POP-Q stage IV)** with a 2 × 3 cm **decubitus ulcer** on the anterior lip of the cervix, a cystocele and a rectocele. The post-void residual was 250 mL and the renal scan shows **bilateral mild hydronephrosis**. **Summary: third-degree uterovaginal prolapse (procidentia) with a decubitus ulcer, urinary retention and hydronephrosis in a grand multipara.** Plan: admit, reduce it and pack it daily with oestrogen-soaked gauze, catheterise, treat UTI, U&E, biopsy the ulcer if it has not healed in 2 weeks, treat the cough, then vaginal hysterectomy with McCall culdoplasty and anterior and posterior repair."

Worked example 1: recording POP-Q

POP-Q is written as a **3 × 3 grid**. All points except **tvl** are measured **on maximum strain**, in cm relative to the **hymen (0)**: above = minus, below = plus.

Example A: a 45-year-old P5 with a bulge on standing.

Aa 0	**Ba +1**	**C −6**
gh 4	**pb 3**	**tvl 9**
Ap −3	**Bp −3**	**D −8**

- The most distal point is **Ba at +1** (within 1 cm of the hymen): **stage II, anterior compartment (cystocele)**.
- The apex is well supported (C −6); the posterior wall is normal.
- **Plan**: PFMT for at least 16 weeks, or a ring pessary; if she chooses surgery, an **anterior colporrhaphy** (no need for hysterectomy, because the apex is well supported).

Example B: the 62-year-old above, tvl 9 cm.

Aa +3	**Ba +7**	**C +8**
gh 6	**pb 2**	**tvl 9**
Ap +3	**Bp +6**	**D +7**

- C is +8, which is at least **tvl − 2 = +7**: **stage IV (procidentia)**, all three compartments.
- A wide hiatus (gh 6) and a short perineal body (pb 2) suggest a **perineal deficiency** (Level III), so add a perineorrhaphy.

> CLINICAL PEARL
> A **large gap between C and D** (the cervix much lower than the posterior fornix) suggests **cervical elongation** rather than true uterine descent. A Manchester repair (which amputates the cervix) suits this.

Worked example 2: the "cured" cystocele that leaks

A 58-year-old with a stage III cystocele reports that her leaking on cough stopped as the bulge got bigger. With the prolapse reduced by the posterior blade of the Sims speculum, a cough produces a jet of urine from the meatus. **Interpretation**: **occult stress incontinence**, masked by urethral kinking. **Plan**: counsel that repair may unmask leakage; consider a continence procedure at the same time or as a second stage, and teach PFMT.

9. Management

Immediate / emergency

Usually there is no emergency, but four situations turn prolapse into one. Start with **ABCDE**: vital signs, pallor, hydration and mental state, because many of these women are elderly and frail.

1. **Irreducible, oedematous, ulcerated or gangrenous procidentia**: admit, bed rest with the foot of the bed raised, analgesia, gentle reduction (after reducing oedema) and a pack (procidentia protocol below).
2. **Acute urinary retention** (a painful palpable bladder, overflow dribbling): **catheterise now** and measure the volume drained.
3. **Signs of obstructive uropathy or sepsis**: fever, loin pain, confusion, reduced urine output, vomiting in an elderly woman with procidentia. Check vitals, RBS, and send **U&E urgently**; start IV fluids and antibiotics for pyelonephritis if septic.
4. **Heavy bleeding** from an ulcer or the cervix: pressure, pack, Hb, and think of **carcinoma**.

Procidentia with decubitus ulcer: step by step

1. **Admit. Bed rest** with the foot of the bed elevated.
2. **Assess**:

- Hb, U&E, **renal ultrasound** (hydronephrosis)
- urinalysis and culture
- **catheterise** if in retention

3. **Reduce the prolapse gently.** Lubricate, and push it back along the vaginal axis. If it is **oedematous**, first apply cold compresses or hypertonic (e.g. magnesium sulphate or glycerine) soaked gauze to reduce the oedema.
4. **Keep it reduced** with a **vaginal pack changed daily**. Use gauze soaked in **oestrogen cream** where available (or glycerine or an antiseptic per the ward's practice). Alternatively, use a **ring pessary** if it will stay in.
5. **Treat infection** (swab if purulent), and treat UTI.
6. **Most decubitus ulcers heal in 1–3 weeks.** If an ulcer **has not healed after about 2 weeks**, or looks suspicious, **biopsy it** to exclude **carcinoma**.
7. Correct anaemia, optimise comorbidities, and treat cough and constipation.
8. **Definitive surgery** once the ulcer has healed and the tissues are healthy. This is usually a **vaginal hysterectomy with vault suspension and repair**, or **colpocleisis** if she is frail.

Definitive

A. Principles

1. **Asymptomatic or mild**: reassurance and lifestyle advice. Prolapse is **not dangerous** and may not progress.
2. **Symptomatic**: the options are **conservative (PFMT, pessary)** or **surgery**, chosen by symptoms, **wish for fertility, sexual activity, fitness** and her preference.
3. **Always**:
   - treat what raises intra-abdominal pressure (cough, constipation, obesity)
   - treat **atrophy** (vaginal oestrogen) and UTI
   - check the kidneys and bladder emptying

B. Conservative management

1. Lifestyle:

- **weight loss** if BMI is over 30
- **treat chronic cough** (stop smoking, reduce smoke exposure, treat TB or COPD)
- **prevent constipation** (fibre, fluids, lactulose)
- **avoid heavy lifting**: advise sharing loads and lifting with the legs

2. Pelvic floor muscle training (PFMT):

- NICE advises a **supervised programme for at least 16 weeks** for symptomatic stage I–II prolapse.
- A typical regimen is **at least 8–12 slow maximal contractions (held up to 10 s), 3 times a day**, plus fast contractions, and the **"knack"** (contract before a cough or lift).
- Teach it by vaginal examination: "squeeze as if holding in wind and urine, lift up and in".
- It improves symptoms and may stop progression. It does not reverse advanced prolapse.

3. Vaginal oestrogen for post-menopausal women with **atrophy**. It improves the epithelium, pessary tolerance and ulcer healing, and prepares the tissues before surgery.

4. Vaginal pessaries:

Type	Use
Ring (with or without support)	First choice for most **stage I–III** uterine or anterior prolapse. The woman may be able to remove and replace it herself, and intercourse is possible.

Type	Use
Gellhorn / shelf	Advanced prolapse, a wide hiatus, or when a ring falls out. Intercourse is not possible. A clinician removes it.
Cube / donut	Advanced prolapse, or when others fail.

Pessary care:

- Fit the **largest size that is comfortable** and that is retained on straining and walking, with space for a finger between the ring and the vaginal wall.
- Make sure she **can void** with it in.
- Review at **2–4 weeks**, then every **3–6 months** to **remove, clean, inspect the vagina** (for erosion) and reinsert.
- Use **vaginal oestrogen** alongside.
- Complications: discharge, ulceration, bleeding, **impaction**, and **fistula if forgotten for years**.

> **UGANDA CONTEXT**
> Before inserting a pessary, make sure she **can return** for regular changes. A neglected pessary can erode into the bladder or rectum. For a rural woman who cannot attend follow-up, **surgery may be the safer option** if she is fit.

C. Surgical management: choosing the operation

Prolapse	Operation	Notes
Uterine prolapse, family complete	**Vaginal hysterectomy + vault support** (McCall culdoplasty or high uterosacral ligament suspension, or sacrospinous fixation) ± anterior or posterior repair	The commonest operation in Uganda. The vault **must** be supported to prevent vault prolapse.
Uterine prolapse, wants to keep the uterus (family complete) **or cervical elongation**	**Manchester (Fothergill) repair**: **amputation of the cervix**, **plication (shortening) of the cardinal ligaments** in front of the cervical stump, and **anterior colporrhaphy** (± posterior repair)	Shorter operation. **Cervical stenosis** (haematometra), **cervical incompetence**, infertility and miscarriage risk, so avoid it if more children are wanted. Screen the cervix first.
Uterine prolapse, wants future fertility	Conservative (PFMT, pessary) until her family is complete, or **sacrohysteropexy** or **sacrospinous hysteropexy** (uterus-preserving)	Delivery after repair: discuss CS.
Cystocele	**Anterior colporrhaphy**: incise the anterior wall, dissect the bladder off, **plicate the pubocervical fascia** in the midline, trim and close the vagina	Native tissue. **NICE: do not use transvaginal mesh** for anterior repair except in research. Recurrence is common (up to about 30%+).
Rectocele / perineal deficiency	**Posterior colporrhaphy + perineorrhaphy**: plicate the rectovaginal fascia and rebuild the perineal body	Avoid **levator plication** in sexually active women (dyspareunia).
Enterocele	Excise or close the sac high (purse-string), **McCall culdoplasty**	Often with hysterectomy or vault surgery.
Vault prolapse	**Sacrospinous ligament fixation** (vaginal: the vault is stitched to the sacrospinous ligament, usually on the right) or **sacrocolpopexy** (abdominal or laparoscopic: mesh from the vault to the sacral promontory)	Sacrocolpopexy is the most durable, but it involves mesh and a laparotomy or laparoscopy. Sacrospinous fixation risks buttock pain and pudendal vessel or nerve injury.

Prolapse	Operation	Notes
Elderly, frail, not sexually active	**Colpocleisis** (LeFort partial or total): closes the vagina	Short, often under regional or local anaesthesia, very high success. **No future intercourse**, so consent carefully. Exclude uterine pathology first (ultrasound, sampling).

D. Peri-operative care (vaginal hysterectomy and repair)

1. **Before surgery**:
   - consent (recurrence, SUI, dyspareunia, bladder or ureteric injury)
   - **treat UTI**
   - **vaginal oestrogen for 4–6 weeks** if atrophic
   - Hb above 10 g/dL ideally
   - VTE risk assessment
   - **spinal anaesthesia** is common
2. **Antibiotic prophylaxis**: a single dose at induction.
3. **During surgery**: good haemostasis, **identify the ureters** mentally (they lie about 1.5–2 cm lateral to the cervix), support the vault, and check bladder integrity (cystoscopy or dye) if in doubt.
4. **After surgery**:
   - **vaginal pack and urinary catheter for about 24 hours** (longer after extensive anterior repair, per the surgeon)
   - after catheter removal, **trial of void** with a residual check (retention is common)
   - early mobilisation
   - thromboprophylaxis
   - stool softeners
5. **Discharge advice**:
   - avoid **heavy lifting and straining for about 6 weeks** (ideally longer)
   - no intercourse for about 6 weeks
   - continue PFMT lifelong
   - treat cough and constipation
   - review at 6 weeks

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Estriol 0.1% vaginal cream (0.5 mg per 0.5 g applicator)	0.5 g	Vaginal	Nightly, then twice weekly	Nightly for 2–3 weeks, then twice weekly; **4–6 weeks** before surgery if atrophic	Atrophy, pessary users, pre-operative preparation, ulcer healing. Minimal systemic absorption
Conjugated oestrogen vaginal cream (alternative)	0.5–1 g	Vaginal	Nightly, then twice weekly	Nightly for 2 weeks, then twice weekly	Use whichever is available
Cefazolin (surgical prophylaxis)	2 g	IV	Single dose within 60 min before incision	Once	Or ceftriaxone 1 g IV **plus metronidazole 500 mg IV** per the local protocol for vaginal hysterectomy

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Enoxaparin (VTE prophylaxis)	40 mg	SC	Once daily	Until mobile	Or mechanical prophylaxis. Many older women have several risk factors; renal function
Nitrofurantoin (UTI)	100 mg	Oral	Every 12 hours	5 days	Adjust to the culture; avoid in renal impairment
Lactulose	15 mL	Oral	Every 12 hours	While at risk of straining	Avoid straining after posterior repair
Paracetamol	1 g	Oral	6-hourly	Post-operative	Max 4 g/day
Diclofenac	50 mg	Oral	8-hourly	Short course	Avoid NSAIDs in renal impairment (obstructive uropathy)
Hypertonic soaks (magnesium sulphate or glycerine gauze)	Topical	To the prolapse	Before reduction	Until oedema settles	For oedematous procidentia (ward practice) Verify *no standard dose in the source notes*

Referral criteria

Level	What to do	Refer up when
HC III	Recognise, counsel, lifestyle advice, teach PFMT, treat UTI, catheterise retention, pregnancy test, VIA	Symptomatic prolapse wanting treatment, procidentia, ulcer, retention, bleeding → HC IV / General Hospital
HC IV / General Hospital	Residual, dipstick, Hb, ultrasound (renal, pelvic), pessary fitting where stocked, admission for reduction and packing	Surgery needed, hydronephrosis with raised creatinine, non-healing ulcer needing histology, suspected cancer → RRH (or a surgical camp)
RRH	Vaginal hysterectomy with vault support, anterior and posterior repair, Manchester repair, colpocleisis, sacrospinous fixation, histology, culture, U&E	Recurrent or vault prolapse needing sacrocolpopexy, urodynamics, persistent obstructive uropathy, carcinoma → NRH
NRH (Mulago) and large PNFP hospitals (Mengo)	Sacrocolpopexy, urodynamics, urology, gynae-oncology	n/a

Monitoring and follow-up

- **Procidentia on the ward**: daily pack change; ulcer healing (biopsy if not healed by about 2 weeks); urine output and creatinine; residual after catheter removal.
- **PFMT**: supervised for at least 16 weeks before judging success; lifelong after surgery.
- **Pessary**: review at **2–4 weeks**, then every **3–6 months** (remove, clean, inspect, reinsert); warning symptoms (bleeding, discharge, pain, retention).
- **After surgery**: trial of void with residual; review at **6 weeks**; watch for recurrence (especially a missed apex), de novo SUI, and dyspareunia.
- **Hydronephrosis**: repeat U&E and renal ultrasound after reduction and surgery.

10. Complications

Disease-related

- **Decubitus ulcer**, bleeding, infection.
- **Urinary retention**, recurrent UTI, **hydronephrosis, renal failure**.
- Irreducibility, oedema, strangulation or gangrene (rare).
- Faecal evacuation problems.
- Sexual dysfunction, social isolation.
- Squamous carcinoma arising in the prolapse (rare).

Treatment-related

Complication	Cause	Recognition and action
Recurrence (especially anterior compartment, and missed apical prolapse)	Anterior repair without apical support; persistent "pushers"	Support the vault; treat cough and constipation
De novo stress incontinence	Unmasked occult SUI	Test before surgery with the prolapse reduced; continence procedure
Urinary retention, UTI	Surgery, catheter	Trial of void with residual
Bladder or ureteric injury, fistula (rare)	Vaginal hysterectomy, anterior repair	Identify the ureters; cystoscopy or dye if in doubt
Haemorrhage, pelvic haematoma, vault infection	Surgery	Haemostasis; prophylactic antibiotics
Dyspareunia, a shortened or narrowed vagina	Posterior repair (levator plication), colpocleisis	Avoid levator plication in sexually active women; consent
Cervical stenosis, incompetence, infertility	Manchester repair	Avoid if more children are wanted
Buttock pain, pudendal vessel injury	Sacrospinous fixation	Counsel
Mesh erosion, bowel injury	Sacrocolpopexy	Counsel; NICE: no transvaginal mesh for anterior or posterior repair outside research
Discharge, erosion, impaction, **fistula** if neglected	Pessary	Regular review

11. Prognosis

- Prolapse is **not dangerous** in most women and may not progress; only a minority of women with prolapse on examination have symptoms.
- **PFMT** improves symptoms and may stop progression in stage I–II, but does not reverse advanced prolapse.
- **Pessaries** control symptoms if follow-up is assured.
- **Native-tissue anterior repair** recurs commonly (up to about 30%+), especially if the apex was missed; **sacrocolpopexy** is the most durable operation for vault prolapse; **colpocleisis** has very high success.
- **Hydronephrosis** usually resolves after reduction and surgery.

12. Prevention

- **Skilled birth attendance**:
  - avoid **premature pushing and fundal pressure**
  - manage a prolonged second stage appropriately
  - repair perineal tears well

- **Postnatal PFMT.**
- **Family planning** (fewer, spaced births).
- Weight control; treat chronic cough (TB, cooking smoke, COPD) and constipation; share heavy loads.
- **Vault support at every hysterectomy** (attach the uterosacral and cardinal ligaments to the vault).

13. Special Populations

- **Young woman who wants more children**: conservative (PFMT, pessary) until her family is complete, or uterus-preserving surgery (sacrohysteropexy, sacrospinous hysteropexy); **no Manchester repair or hysterectomy**; discuss CS for later deliveries.
- **Young nulliparous woman**: look for spina bifida occulta, neurological disease or connective tissue disorder.
- **Pregnancy** (rare): risk of cervical oedema, infection and obstructed labour.
- **Elderly, frail, not sexually active**: pessary or colpocleisis (after excluding uterine pathology).
- **Post-menopausal with atrophy**: vaginal oestrogen before a pessary or surgery.
- **Rural woman who cannot return for follow-up**: surgery may be safer than a pessary if she is fit.
- **Renal impairment from obstructive uropathy**: avoid NSAIDs and nitrofurantoin; reduce and drain first.

14. Key Points

High-yield points

- Treat symptoms, not the grade.
- Sims speculum, left lateral, woman straining; rectovaginal exam for an enterocele.
- POP-Q: the hymen is zero; record a 3 × 3 grid.
- Missed apical prolapse is the main cause of recurrence: support the vault.
- Check the residual, U&E and kidneys in every advanced prolapse.
- Reduce and cough: occult SUI.
- Biopsy any ulcer not healed after about 2 weeks.
- PFMT for at least 16 weeks first; pessary only if follow-up is assured.
- Three questions choose the operation: more children? sexually active? fit for anaesthesia?
- No transvaginal mesh for anterior or posterior repair outside research.

Numbers & doses

Item	Value
POP-Q reference point	The **hymen = 0**; above minus, below plus; all at **maximum strain** except tvl
Aa / Ap	**3 cm** above the hymen; range −3 to +3
Stage I / II / III / IV	Leading edge **< −1** / **−1 to +1** / **> +1 but < tvl − 2** / **≥ tvl − 2** (complete eversion)
Post-void residual suggesting incomplete emptying	**Over about 100–150 mL**
Supervised PFMT before judging success (NICE)	**At least 16 weeks**
PFMT regimen	**8–12 slow maximal contractions** (held up to 10 s), **3 times a day**, plus fast contractions and "the knack"
Pessary review	**2–4 weeks** after fitting, then every **3–6 months**

Item	Value
Decubitus ulcer	Most heal in **1–3 weeks** with reduction and daily packing; **biopsy if not healed after about 2 weeks**
Vaginal oestrogen before surgery	**4–6 weeks** if atrophic; estriol 0.1% cream **0.5 g nightly for 2–3 weeks, then twice weekly**
Hb before surgery	Ideally **above 10 g/dL**
Ureter distance from the cervix	About **1.5–2 cm** lateral
Surgical prophylaxis	Cefazolin **2 g IV** within 60 min of incision (or ceftriaxone 1 g + metronidazole 500 mg IV)
VTE prophylaxis	Enoxaparin **40 mg SC daily** until mobile
Pack and catheter after vaginal hysterectomy/repair	About **24 hours** (longer after extensive anterior repair)
After surgery	No heavy lifting or intercourse for about **6 weeks**; review at **6 weeks**
Anterior repair recurrence	Up to about **30%+**
Weight loss advised	If **BMI over 30**
Uterine sound length suggesting elongation	Much over **8 cm** (mostly cervix)

Exam traps

- **Using a Cusco speculum.** It hides the walls; use a **Sims speculum in the left lateral position** with her straining.
- **Operating on the grade, not the symptoms.** Asymptomatic stage II needs no surgery.
- **Doing an anterior repair alone when the apex is down.** Missed apical prolapse is the main cause of recurrence: support the vault.
- **Calling every ulcer "decubitus".** Biopsy any ulcer that is suspicious or has not healed after about 2 weeks: it may be carcinoma.
- **Forgetting the kidneys and residual** in procidentia (hydronephrosis, retention, renal failure).
- **Manchester repair for a young woman who wants more children.** No: cervical stenosis and incompetence; use conservative or uterus-preserving options.
- **Promising dryness after repair.** Test for **occult SUI** with the prolapse reduced first.
- **Colpocleisis without checking sexual wishes and the endometrium.** No future intercourse; exclude uterine pathology first.
- **Calling every vaginal mass a prolapse.** Fibroid polyp (ring of cervix around the stalk) and chronic inversion (no os visible) are mimics.

Questions seniors ask

On the ward round

- **How do you record POP-Q?** Nine measurements in a 3 × 3 grid (Aa, Ba, C / gh, pb, tvl / Ap, Bp, D), in cm relative to the hymen on maximum strain, tvl at rest; the stage comes from the most distal point.
- **How do you tell an enterocele from a rectocele?** Rectovaginal examination: bowel between the fingers above the rectum = enterocele; the rectum bulging into the vagina = rectocele.
- **What investigations before surgery for procidentia?** Urine culture, U&E, renal ultrasound, post-void residual, FBC and cross-match, cervical screening, pelvic ultrasound, and biopsy of any non-healing ulcer.
- **What three questions choose the operation?** Does she want more children? Is she sexually active? Is she fit for anaesthesia?

- **How do you manage a decubitus ulcer?** Admit, reduce, daily pack with oestrogen cream, treat infection, biopsy if not healed in about 2 weeks, then definitive surgery.
- **How do you fit a pessary safely?** The largest comfortable size, retained on straining and walking, a finger's space, able to void; review in 2–4 weeks then every 3–6 months with vaginal oestrogen.
- **How do you suspect cervical elongation?** The fundus is at its normal level on bimanual examination, the cervix is long, and on POP-Q point C is much lower than point D.

Viva questions

1. **What are DeLancey's levels?** I = the uterosacral–cardinal complex (apex). II = paracolpium to the arcus tendineus (mid-vagina). III = the perineal body and membrane (distal). Level I failure causes uterine or vault prolapse.
2. **What does the Manchester repair involve, and what are its disadvantages?** Cervical amputation, cardinal ligament plication and anterior repair. Disadvantages: cervical stenosis or incompetence, infertility, miscarriage, and difficult later endometrial assessment.
3. **Which operations treat vault prolapse?** Sacrospinous fixation (vaginal) or sacrocolpopexy (abdominal or laparoscopic with mesh). Colpocleisis if she is frail.
4. **How do you tell an enterocele from a rectocele?** Rectovaginal examination: bowel between the fingers above the rectum indicates an enterocele. It is also seen at the upper posterior wall or vault.
5. **Why does procidentia cause hydronephrosis?** The ureters are dragged down through the levator hiatus and kinked or compressed.
6. **What is first-line treatment for symptomatic mild prolapse?** Supervised PFMT for at least 16 weeks, plus lifestyle changes. Then a pessary or surgery.
7. **What is occult SUI?** SUI masked by urethral kinking from the prolapse. It appears after repair, so test with the prolapse reduced.

Mnemonics

- **Risk factors, "PELVIC SAG"**: **P**arity (vaginal), **E**lderly, **L**ifting heavy loads, **V**aginal instrumental delivery / large babies, **I**ntra-abdominal pressure (cough, constipation, obesity, ascites), **C**onnective tissue disorders, **S**urgery (previous hysterectomy), **A**trophy (menopause), **G**enetic / neurological (spina bifida).
- **Front, top, back**: **F**ront = bladder (cystocele), **T**op = uterus, vault or enterocele, **B**ack = rectum (rectocele).
- **Manchester = "ACP"**: **A**mputate the cervix, **C**ardinal ligaments plicated, **P**lus anterior repair.
- **POP-Q: "the hymen is zero"**. Above is minus, below is plus. Stage II is ±1 around the hymen.
- **DeLancey: "Suspend, Attach, Fuse"**. Levels I, II and III.

OSCE tips

- For **PFMT teaching**, use simple language, check understanding by asking her to demonstrate, and give a daily routine.
- For the **pessary** station: explain the options, check the fit (a finger's space, retained on cough and standing, voiding possible), follow-up every 3–6 months, and warning symptoms (bleeding, discharge, pain, retention).
- For **examination**: consent and a chaperone, **Sims speculum in the left lateral position**, **ask her to strain**, and check each compartment in turn; reduce and ask her to cough; finish with a rectovaginal examination and the residual.

15. References

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define pelvic organ prolapse.	Descent of one or more of the anterior vaginal wall, posterior vaginal wall, uterus (cervix) or vaginal vault (after hysterectomy) into or through the vagina, with the bladder, rectum or bowel behind them.
2	DeLancey level I: structures, and what failure causes?	The **uterosacral–cardinal ligament complex**, which suspends the apex. Failure causes **uterine or vault prolapse and enterocele**.
3	DeLancey level II and III?	**II**: the paracolpium attaches the mid-vagina to the **arcus tendineus fasciae pelvis** (failure causes cystocele or rectocele). **III**: the distal vagina is fused to the **perineal body and perineal membrane** (failure causes urethrocele, low rectocele, gaping introitus).
4	Strongest risk factor for prolapse?	**Vaginal childbirth / parity**, especially large babies, a prolonged second stage and instrumental delivery. These cause levator avulsion, pudendal nerve injury and fascial tears.
5	POP-Q staging (stages 0–IV)?	The hymen = 0. **I**: leading edge more than 1 cm above the hymen. **II**: within 1 cm above or below the hymen. **III**: more than 1 cm below the hymen but less than tvl – 2 cm. **IV**: complete eversion.
6	POP-Q points?	**Aa, Ba** (anterior), **C** (cervix/vault), **D** (posterior fornix), **Ap, Bp** (posterior), **gh** (genital hiatus), **pb** (perineal body), **tvl** (total vaginal length). Aa and Ap sit 3 cm inside the hymen (range −3 to +3).
7	Traditional degrees of uterine prolapse?	**1st**: the cervix descends within the vagina. **2nd**: the cervix reaches or protrudes through the introitus on straining. **3rd (procidentia)**: the whole uterus lies outside the introitus.
8	Which speculum and position for prolapse assessment?	**Sims speculum, left lateral position**, with the woman **straining**. Retract the posterior wall to see the anterior wall (cystocele) and the cervix. Retract the anterior wall to see the posterior wall (rectocele or enterocele).
9	How do you distinguish an enterocele from a rectocele?	**Rectovaginal examination**. A rectocele is the rectal wall bulging, felt with the rectal finger. An **enterocele** has small bowel between the vaginal and rectal fingers, higher up (upper posterior wall or vault).
10	What is occult stress incontinence?	SUI masked by urethral kinking from a cystocele. It is revealed by a **cough test with the prolapse reduced**, and it may appear after repair.
11	First-line treatment for symptomatic mild–moderate prolapse (NICE)?	**Supervised pelvic floor muscle training for at least 16 weeks**, plus lifestyle: weight loss, treat cough and constipation, avoid heavy lifting, vaginal oestrogen if atrophic.
12	Types of pessary and their uses?	**Ring** (± support): stage I–III; she can self-manage; intercourse is possible. **Gellhorn / shelf / cube / donut**: advanced prolapse or a wide hiatus. Review every 3–6 months: remove, clean, inspect for erosion.
13	Components of the Manchester (Fothergill) repair?	**Amputation of the cervix**, **plication of the cardinal ligaments** in front of the cervical stump, and **anterior colporrhaphy** (± posterior repair). It preserves the uterus.
14	Disadvantages of the Manchester repair?	**Cervical stenosis** (haematometra, difficult endometrial sampling), **cervical incompetence**, infertility and miscarriage. Avoid if she wants more children.

#	Question	Answer
15	Operations for vault prolapse?	**Sacrospinous ligament fixation** (vaginal) or **sacrocolpopexy** (abdominal or laparoscopic, with mesh to the sacral promontory; the most durable). **Colpocleisis** if frail and not sexually active.
16	What is McCall culdoplasty and why do it?	At vaginal hysterectomy, the uterosacral ligaments are plicated and attached to the vault, obliterating the cul-de-sac. It **supports the vault** (level I) and prevents enterocele and vault prolapse.
17	Management of procidentia with a decubitus ulcer?	Admit, bed rest, reduce the prolapse, keep it reduced with a **daily vaginal pack** (oestrogen cream where available), treat UTI, check U&E and renal US, **biopsy if not healed at about 2 weeks**, then definitive surgery.
18	Why can procidentia cause renal failure?	The ureters are dragged down through the levator hiatus and **kinked or compressed**, causing bilateral **hydronephrosis** and obstructive uropathy.
19	NICE position on transvaginal mesh?	Do **not** use transvaginal mesh for anterior or posterior wall prolapse repair except in a research context, because of erosion, pain and dyspareunia. Native-tissue repair is standard.
20	What is colpocleisis and who is it for?	Surgical **closure of the vagina** (LeFort partial or total). It is for **elderly, frail women who are not sexually active**. It has a very high success rate and a short operation time. Exclude uterine pathology first.
21	How is a POP-Q examination recorded?	A **3 × 3 grid**: **Aa | Ba | C** / **gh | pb | tvl** / **Ap | Bp | D**, in cm relative to the **hymen (0)**; above = minus, below = plus. All points on **maximum strain** except **tvl (at rest)**. The stage comes from the most distal point.
22	POP-Q: Aa 0, Ba +1, C −6, gh 4, pb 3, tvl 9, Ap −3, Bp −3, D −8. Stage and compartment?	Leading edge **Ba +1** (within 1 cm of the hymen) = **stage II anterior (cystocele)**; apex and posterior wall well supported. Options: PFMT for at least 16 weeks, ring pessary, or anterior colporrhaphy.
23	Which bedside tests do you do in a woman with procidentia?	**Post-void residual** (over about 100–150 mL = incomplete emptying), **urine dipstick** (then culture), **cough test with the prolapse reduced** (occult SUI), Hb, RBS, VIA, and pregnancy test if premenopausal; then U&E and **renal ultrasound** for hydronephrosis.
24	What four situations make prolapse an urgent admission?	**Irreducible, oedematous or gangrenous procidentia**; **acute urinary retention**; **obstructive uropathy or urosepsis** (fever, loin pain, rising creatinine); **bleeding from an ulcer** (think carcinoma).
25	On POP-Q, what suggests cervical elongation rather than true uterine descent?	A **large gap between point C (cervix) and point D (posterior fornix)**: the cervix is low but the fornix and fundus stay high. Confirm on bimanual exam (fundus at its normal level). A **Manchester repair** suits this.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 58-year-old para 6, post-menopausal, complains of a vaginal bulge for 2 years. It is worse in the evening, and she sometimes has to push it back to pass urine. On straining, the cervix reaches 2 cm beyond the hymen. What is the POP-Q stage?

- A Stage 0
- B Stage I
- C Stage II
- D Stage III
- E Stage IV

2. A 32-year-old para 3 with symptomatic stage II uterine prolapse wants two more children. What is the MOST appropriate management now?

- A Vaginal hysterectomy with McCall culdoplasty
- B Supervised pelvic floor muscle training and/or a ring pessary, deferring surgery until her family is complete
- C Manchester repair
- D LeFort colpocleisis
- E Sacrocolpopexy

3. A 70-year-old para 9 presents with complete procidentia that has been out for months, a 3 cm bleeding ulcer on the cervix, and a creatinine of 180 µmol/L. What is the MOST appropriate initial management?

- A Immediate vaginal hysterectomy
- B Discharge with a ring pessary and review in 6 months
- C Radiotherapy for presumed cervical cancer
- D Colpocleisis under local anaesthesia today
- E Admit, reduce the prolapse and pack daily, catheterise, treat infection, scan the kidneys, and biopsy the ulcer if it does not heal

4. On a rectovaginal examination of a woman with a posterior wall bulge, loops of bowel are felt between the vaginal and rectal fingers high in the rectovaginal septum. What is the diagnosis?

- A Enterocele
- B Rectocele
- C Cystocele
- D Urethrocele
- E Rectal prolapse

5. A 55-year-old is booked for anterior repair for a large cystocele. She denies incontinence. What pre-operative test is MOST important to avoid a new post-operative leakage problem?

- A Serum oestradiol
- B Pelvic MRI
- C A cough stress test with the prolapse reduced
- D Colonoscopy
- E Cystoscopy

6. A 45-year-old para 5 with a very long cervix (uterine sound 11 cm, fundus at its normal level) and a cystocele has completed her family and wants to keep her uterus. Which operation BEST suits her?

A Sacrospinous fixation
B Colpocleisis
C Posterior colporrhaphy alone
D Manchester (Fothergill) repair
E Sacrocolpopexy

7. Two years after an abdominal hysterectomy, a 60-year-old sexually active woman has complete vault eversion. Which is the MOST durable surgical option?

A Abdominal (or laparoscopic) sacrocolpopexy
B Anterior colporrhaphy
C Manchester repair
D Ring pessary
E LeFort colpocleisis

8. An 82-year-old widow with heart failure and COPD has symptomatic stage IV procidentia. She is not sexually active and a pessary repeatedly falls out. Which surgical option is MOST appropriate?

A Abdominal sacrohysteropexy under general anaesthesia
B Colpocleisis under regional anaesthesia
C Laparoscopic sacrocolpopexy
D Manchester repair
E No treatment is possible

9. A 50-year-old farmer has a symptomatic stage II cystocele. She is overweight (BMI 33), has a chronic smoker's cough and constipation, and prefers to avoid surgery. What is the BEST initial plan?

A Anterior repair with transvaginal mesh
B Vaginal hysterectomy
C Reassure her; no action is needed
D Manchester repair
E Supervised PFMT for at least 16 weeks, weight loss, stopping smoking, treating the cough and constipation, with a pessary if symptoms persist

10. A pessary user, reviewed after being lost to follow-up for 3 years, reports a foul discharge and continuous leakage of urine. What is the MOST likely complication?

A Bacterial vaginosis only
B Overactive bladder
C Pessary erosion with a vesicovaginal fistula
D Detrusor underactivity
E Stress urinary incontinence

11. A 65-year-old had an anterior colporrhaphy alone for a 'cystocele' 1 year ago. She now has a recurrent bulge, and the cervix descends to the hymen. What is the MOST likely reason for the failure?

A Absorbable sutures were used
B Unrecognised apical (uterine) prolapse, a level I defect not addressed
C Post-operative UTI
D Excess vaginal oestrogen
E Inadequate bowel preparation

12. A 16-year-old nulliparous girl presents with uterine prolapse to the introitus. What should you look for FIRST?

A History of instrumental delivery
B Menopausal status
C Chronic uterine inversion
D Neurological cause (e.g. spina bifida occulta) or a connective tissue disorder
E Ovarian cancer

13. During vaginal hysterectomy for uterine prolapse, what step MOST reduces the risk of subsequent vault prolapse?

A Attaching the uterosacral–cardinal ligaments to the vault / McCall culdoplasty
B Leaving the vault open
C Removing both ovaries
D Inserting a transvaginal mesh
E Excising a large amount of vaginal skin

14. A mass protrudes from the vagina of a 40-year-old with heavy periods. On speculum examination a firm, bleeding mass emerges through the os, with a rim of cervix felt around its stalk. The uterus is bulky. What is the MOST likely diagnosis?

A Third-degree uterine prolapse
B Chronic uterine inversion
C Cervical elongation
D Gartner's duct cyst
E Fibroid polyp (prolapsed pedunculated submucous fibroid)

15. The day after vaginal hysterectomy and anterior repair, the catheter is removed. Six hours later she has lower abdominal discomfort and has passed only small amounts of urine. A bladder scan shows 650 mL. What is the MOST appropriate action?

A Give furosemide
B Encourage oral fluids and review tomorrow
C Re-catheterise (indwelling for 24–48 hours or intermittent), then repeat the trial of void, and check for UTI
D Return to theatre for re-exploration
E Start oxybutynin

Answers & explanations

> **1. D** · Stage III
>
> At **+2 cm** (more than 1 cm below the hymen) but not complete eversion, this is **stage III**. **A** Stage 0 means no prolapse. **B** Stage I is more than 1 cm above the hymen. **C** Stage II is within ±1 cm of the hymen. **E** Stage IV is complete eversion (at least tvl – 2).

> **2. B** · Supervised pelvic floor muscle training and/or a ring pessary, deferring surgery until her family is complete
>
> A woman wanting future pregnancies should be managed **conservatively (PFMT, pessary)** until her family is complete, or with uterus-preserving hysteropexy in selected cases. **A** removes her fertility. **C** Manchester causes cervical incompetence or stenosis and poor fertility outcomes. **D** closes the vagina. **E** is for vault prolapse (after hysterectomy).

3. E · Admit, reduce the prolapse and pack daily, catheterise, treat infection, scan the kidneys, and biopsy the ulcer if it does not heal

Decubitus ulcers usually heal with **reduction and daily packing**, which also relieves the ureteric kinking (hydronephrosis, raised creatinine). Biopsy an ulcer that persists, then plan surgery. **A** Operating through an infected ulcer with renal impairment is unsafe. **B** She needs admission and assessment. **C** Cancer is not yet proven, so biopsy first. **D** Surgery before ulcer healing and renal assessment is premature.

4. A · Enterocele

An **enterocele** (a hernia of the pouch of Douglas) places **small bowel between the fingers**, high on the posterior wall. **B** A rectocele is the rectum itself bulging lower down. **C/D** are anterior compartment. **E** Rectal prolapse protrudes through the anus.

5. C · A cough stress test with the prolapse reduced

A large cystocele can **kink the urethra and mask stress incontinence (occult SUI)**. Reducing the prolapse and asking her to cough reveals it, so it can be counselled for or treated. **A/B/D/E** do not address occult SUI.

6. D · Manchester (Fothergill) repair

Cervical elongation with a cystocele, a completed family and a wish to keep the uterus is the classic indication for the **Manchester repair** (cervical amputation, cardinal ligament plication, anterior repair). **A/E** are vault procedures. **B** closes the vagina, which is inappropriate at 45 unless she requests it. **C** doesn't address the cervix or cystocele.

7. A · Abdominal (or laparoscopic) sacrocolpopexy

Sacrocolpopexy (mesh from the vault to the sacral promontory) has the best long-term anatomical success for vault prolapse, and it preserves vaginal length for intercourse. Sacrospinous fixation is the vaginal alternative. **B** doesn't support the apex. **C** requires a cervix. **D** is conservative, not surgery. **E** stops intercourse, which she wants to continue.

8. B · Colpocleisis under regional anaesthesia

Colpocleisis is quick, can be done under regional or local anaesthesia, and has very high success in frail women who are not sexually active. **A/C** involve longer abdominal surgery under GA in a high-risk patient. **D** is less suitable for procidentia in the elderly. **E** is untrue.

9. E · Supervised PFMT for at least 16 weeks, weight loss, stopping smoking, treating the cough and constipation, with a pessary if symptoms persist

NICE recommends **supervised PFMT for at least 16 weeks** plus **lifestyle modification** as first line for symptomatic stage I–II prolapse, with a pessary as the next option. **A** NICE advises against transvaginal mesh outside research. **B/D** are surgery she wishes to avoid, without trying conservative care. **C** She is symptomatic, so she needs treatment.

10. C · Pessary erosion with a vesicovaginal fistula

A **neglected pessary** can erode through the vaginal wall into the bladder (**VVF**) or rectum, giving foul discharge and **continuous** urinary leakage. **A** BV does not cause continuous urine leakage. **B/E** cause urge or effort leakage, not continuous. **D** causes retention.

11. B · Unrecognised apical (uterine) prolapse, a level I defect not addressed

Missed apical prolapse is the main cause of early 'recurrence' after an anterior repair. The level I (uterosacral–cardinal) defect must be treated too. **A** Absorbable sutures are standard. **C/D/E** don't cause apical descent.

12. D · Neurological cause (e.g. spina bifida occulta) or a connective tissue disorder

Prolapse in a **nulliparous young woman** suggests **congenital weakness**: neurological causes (spina bifida, which denervates the pelvic floor) or connective tissue disorders (Ehlers–Danlos, Marfan). Examine the back and lower limbs. **A/B** are not applicable. **C** requires a prior pregnancy. **E** is rare, and a mass effect would be felt.

13. A · Attaching the uterosacral–cardinal ligaments to the vault / McCall culdoplasty

Restoring **level I support** by attaching the uterosacral–cardinal complex to the vault (McCall culdoplasty or high uterosacral suspension) prevents vault prolapse and enterocele. **B/C** have no supportive role. **D** NICE advises against mesh. **E** shortens and narrows the vagina without supporting it.

14. E · Fibroid polyp (prolapsed pedunculated submucous fibroid)

A firm mass through the os with the **cervix felt as a ring around its stalk**, plus menorrhagia and a bulky uterus, is a **fibroid polyp**. **A** In prolapse, the cervix itself (with its os) is the leading part. **B** Inversion has no visible os, and the fundus is absent abdominally. **C** Elongation shows a normal os on a long cervix. **D** Gartner's cysts are in the vaginal wall, not through the os.

15. C · Re-catheterise (indwelling for 24–48 hours or intermittent), then repeat the trial of void, and check for UTI

Post-operative urinary retention is common after anterior repair (oedema, pain). Treat it with **catheterisation** and a repeat trial of void, and check for UTI. **A/B** worsen distension, which can damage the detrusor. **D** isn't indicated for simple retention. **E** Anticholinergics worsen retention.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Procidentia with a bleeding ulcer

A 68-year-old para 10 from a rural village in Mukono is brought by her daughter to Mengo gynaecology clinic with 'the womb coming out' for 5 years. It has stayed out for 8 months, and it now bleeds and smells. She passes urine with difficulty.

1. What focused history do you take?

- Duration, and whether she can push it back.
- **Urinary**: retention, need to reduce to void, UTI, loin pain.
- **Bowel**: constipation, splinting.
- Bleeding or discharge.
- Postmenopausal bleeding.
- Sexual activity.
- Parity and delivery history.
- **Chronic cough**, heavy work.
- Comorbidities (hypertension, diabetes, heart or lung disease).
- Previous surgery.
- Her own goals and her support at home.

2. Examination shows a complete irreducible procidentia with an oedematous everted vagina and a 3 × 3 cm ulcer on the anterior cervix. The bladder is palpable. What do you do first?

1. **Catheterise**, noting the residual.
2. Admit, with bed rest and the foot of the bed raised.
3. **Reduce the oedema** with cold or hypertonic soaks, then **reduce the prolapse** gently.
4. **Daily vaginal pack** (oestrogen cream-soaked if available).
5. Swab and treat infection.
6. Investigations: FBC, U&E, RBS, HIV, urinalysis and culture, **renal ultrasound**, pelvic ultrasound, and a chest assessment.

3. The renal scan shows bilateral moderate hydronephrosis and a creatinine of 160 µmol/L. Why, and what do you expect?

The ureters are **kinked and compressed** by the prolapse through the levator hiatus, causing obstructive uropathy. After **reduction and catheter drainage**, the hydronephrosis and creatinine usually improve. Recheck the U&E and ultrasound. Adjust drug doses (e.g. avoid NSAIDs and nitrofurantoin while the eGFR is low).

4. After 2 weeks the ulcer has not healed and looks raised and irregular. What next?

Biopsy the ulcer to exclude **squamous cell carcinoma** of the cervix or vagina. If malignant, stage it and refer to the gynae-oncology / cancer institute pathway. If benign (decubitus), continue packing until it heals.

5. The biopsy is benign and the ulcer heals. What definitive treatment do you offer?

> If she is fit: **vaginal hysterectomy with McCall culdoplasty or uterosacral vault suspension, plus anterior and posterior colporrhaphy**, under spinal anaesthesia, with antibiotic and VTE prophylaxis. If she is frail and not sexually active: **colpocleisis**. Post-op: catheter and pack for 24 hours, trial of void, and treat the cough and constipation. No heavy lifting for 6+ weeks. PFMT.

> **Take-away:** Before operating on procidentia: reduce it and pack it until the ulcer heals, relieve the retention, check the kidneys, and biopsy any ulcer that won't heal.

Case 2: A young mother with a bulge

A 29-year-old para 3 (last delivery 8 months ago, a 4.1 kg baby after a 3-hour second stage at an HC III) reports a feeling of 'something at the opening', worse after carrying water, plus leakage of urine when she coughs. She wants one more child.

1. What will you examine for, and how?

> - Abdomen.
> - Vulva at rest and while straining.
> - **Cough test** for SUI.
> - **Sims speculum in the left lateral position** with straining: anterior, apical and posterior compartments.
> - **POP-Q**.
> - Bimanual exam.
> - Rectovaginal exam.
> - **Assess pelvic floor contraction** (the Oxford scale).
> - Look at the perineal body for an old tear.

2. She has a stage II cystocele (Ba −0.5), the cervix at −4 and a weak pelvic floor contraction. What is your management plan?

> **Conservative**, because she wants more children:
>
> - **Supervised PFMT for at least 16 weeks** (8–12 slow maximal contractions 3 times daily plus fast contractions, and the 'knack' before a cough or lift).
> - **Lifestyle**: avoid heavy loads, prevent constipation, weight.
> - A **ring pessary** if she remains symptomatic.
> - **Family planning** to space births.
> - Defer surgery until her family is complete.

3. How would you plan her next delivery?

> - Deliver in a **facility with a skilled attendant**, avoid a prolonged second stage and fundal pressure, and repair any tears well.
> - Vaginal birth is usually reasonable for mild prolapse. Discuss the risk of worsening.
> - Postnatal PFMT.
> - If she has had a previous prolapse **repair**, discuss CS.

4. What are you teaching her about why it happened?

> Stretching or tearing of the **levator ani**, fascia and pudendal nerve during a long delivery of a big baby **opened the levator hiatus** and weakened the **level II** (cystocele) support. Carrying heavy loads and straining keep pushing down. Strengthening the muscles and reducing strain can stop it getting worse.

Take-away: A young woman who wants more children needs conservative treatment: PFMT, lifestyle changes, a pessary and family planning. Surgery waits until her family is complete.

Case 3: Recurrence after hysterectomy

A 61-year-old who had an abdominal hysterectomy for fibroids 6 years ago presents with a vaginal lump, difficulty emptying her bowel (she presses on the vagina to defaecate) and dyspareunia. She is sexually active and fit.

1. What compartments are likely involved, and why?

> **Vault prolapse** (a level I defect, possibly because the vault was not attached to the uterosacral–cardinal ligaments at hysterectomy), probably with an **enterocele** and/or **rectocele** (splinting to defaecate suggests a posterior compartment defect).

2. How will you confirm this on examination?

> - **Sims speculum** with straining to watch the vault descend.
> - **POP-Q** point C.
> - **Rectovaginal exam**: small bowel between the fingers means an enterocele; the rectum bulging means a rectocele.
> - Assess the perineal body and the anal sphincter.
> - Cough test for SUI.

3. What are her surgical options, with their pros and cons?

> **1. Sacrospinous fixation (vaginal) ± posterior repair.**
>
> - Pros: vaginal, avoids mesh, shorter recovery.
> - Cons: buttock pain, pudendal injury, anterior recurrence.
>
> **2. Abdominal or laparoscopic sacrocolpopexy.**
>
> - Pros: most durable, preserves vaginal length and axis for intercourse.
> - Cons: mesh erosion, bowel or vascular injury, a longer operation.
>
> **3. Colpocleisis.** Not suitable, because she is sexually active.

4. What peri-operative measures reduce her risk of another recurrence?

> - Treat **constipation** (fibre, lactulose) and **cough**.
> - Weight control.
> - **Avoid heavy lifting**.
> - Vaginal oestrogen if atrophic.
> - **Lifelong PFMT**.
> - Correct technique: address all compartments at surgery.

Take-away: Every hysterectomy must include vault support. When prolapse recurs, identify every compartment involved, and choose the operation to suit her sexual function and fitness.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.